

Christian B. Ibañez

Data Scientist · Operations Research · AI/ML

christianbiongibanez@gmail.com · linkedin.com/in/ibanezchristian · github.com/Cbanzaime23

· cbanzaime23.github.io/Christian-sCV

PROFESSIONAL EXPERIENCE

Data Scientist / AI Solutions Developer

2024 – Present

Sun Life Global Solutions Philippines · BGC, Taguig City

- **Queueing Theory Capacity Analysis:** Formulated and solved analytical queueing models (M/M/c and multi-server variants) to evaluate client-advisor arrival and service distributions in a real-world Client Service Centre, predicting branch utilization rates and deriving optimum staffing configurations. This analysis was used to assess the feasibility of installing Queueing Systems for Sun Life Client Service Center branches.
- **Skills Allocation Semantic Model:** Engineered a semantic mapping system using a BERT transformer model (via Hugging Face) to extract skill embeddings, correlating cosine similarity scores across employee skillset profiles to accelerate internal resource allocation.
- **FTE vs. Chargeable Hours Analysis:** Conducted a rigorous statistical capacity analysis for the Testing COE, applying hypothesis testing to debunk leadership's efficiency hypothesis by demonstrating that rising chargeable hours required proportional staffing adjustments rather than static FTE counts.
- **RAG AI Agent Knowledge Pipeline:** Led the development of a Python-based ETL pipeline that extracts, parses, and shards Planview data for retrieval-augmented generation (RAG) for MS365 Copilot Agent and custom BMAD-style persona Agent. Automates project status analysis and insights generation for PMO Delivery Excellence Team at a Portfolio Level.
- **PMO Project Status Compliance:** Automated the entire weekly PMO project status compliance from data extraction, processing, AI analysis of project status data to identify compliance, and report creation to stakeholders.
- **PCQA Automation:** Worked with PCQA team on end-to-end automation of project artifacts audit, including 2 AI Agents (proactive artifact completeness checking and project artifact audit) and a dashboard for audit results.
- **Batch Job Data Pipeline Orchestration:** Engineered pipeline using Python, FastAPI, PowerShell, and batch scripts integrated with Power Automate to trigger real-time Outlook and MS Teams alerts for batch run errors with automated AI-generated error analysis.
- **BI Dashboard Quality Audit:** Conducted qualitative user interviews and usage-experience logging to diagnose business-unit Tableau dashboards for unrealized business savings, mapping data-quality gaps and SLA mismatches to deliver actionable redesign roadmaps.
- **Agile Footprint Dashboard:** Migrated manual Agile Footprint reporting by architecting a Power Query data model that cleans and merges multi-source sprint records, deploying a Tableau dashboard that reduced monthly report generation from 32 hours to 30 minutes (98% time saved) with 95% data accuracy.
- **Data Literacy Workshops:** Conducted Power Query workshops to cultivate a data-driven mindset and improve data quality awareness, standardizing Excel workflows across PMO, PCQA, and Agile teams.

Trainings:

- Project Management Bootcamp with Generative AI Application
- 7 Habits of Highly Effective People
- Excel Power Query Beginner to Advanced (Udemy)
- Project Management Fundamentals (PMI)
- Data Storytelling for Business (StoryIQ)

Data Scientist / Business Analytics Manager

2021 – 2024

Teleperformance Philippines • EDSA Greenfield District, Mandaluyong City

- **Workforce Agent Scheduling Optimization:** Formulated and prototyped a workforce scheduling optimization model for an insurance account, modeling agent shift schedules, shrinkage patterns, and demand forecasts as a constrained optimization problem to minimize over- and under-capacity risk on a weekly planning horizon. Explored integer programming formulations with scheduling and resource allocation constraints.
- **Predictive Attrition Early Warning System:** Developed and deployed a batch prediction system utilizing supervised classification models (Random Forest/Gradient Boosting) across Healthcare, BFSI, and Travel accounts, achieving a 2–3% monthly attrition reduction; integrated SHAP/LIME explainable AI (XAI) to generate personalized, feature-level coaching recommendations ("who to coach and what to coach") for team leaders, yielding **€2M+ in financially validated savings** in the BFSI sector.
- **Sales Retention Scoring Model:** Engineered and deployed multiple classification scoring models for different product categories (utilizing Python, XGBoost, and PyCaret/AutoML for ensembling) for a Healthcare and Life Sciences client, delivering a **5% revenue increase (~₱2M)** within three months of deployment, validated via pre-and-post-launch business metric tracking.
- **Repeat Chat Root-Cause Analysis:** Conducted statistical hypothesis testing (Chi-Square/ANOVA) and exploratory data analysis in Python to isolate statistically significant drivers of customer contacts; designed a Power BI dashboard fueled by a custom Python-to-SQL preprocessing pipeline to enable proactive, manual tracking of repeat transaction root causes.

Trainings:

- COPC Certified Lean Six Sigma Yellow Belt (2022)
- Attended multiple Design Thinking Workshops

Data Analysis & Visualization Analyst

2018 – 2021

PricewaterhouseCoopers Service Delivery Centre • Rockwell Business Centre, Pasig City

- **Demand & FTE Capacity Optimization Model:** Formulated a Linear/Integer Programming (LP/IP) optimization model to minimize staffing overhead while satisfying fluctuating demand; implemented the model in Excel Solver and engineered an interactive VBA dashboard interface enabling operations Workforce Planners to simulate staffing scenarios, input capacity buffers, and plot optimal schedules.
- **Productivity Report Digitization Project:** Served as Lead Developer, engineering manually-triggered Alteryx ETL workflows to ingest, clean, and join data from 10+ disparate operational databases; designed an interactive Power BI dashboard utilizing advanced DAX calculations and custom data modeling to visualize operational KPI trends, reducing daily preparation time from 3 hours to less than 20 minutes.
- **Asset Management Process Automation:** Designed and implemented a hybrid, multi-tool automation workflow (Excel VBA, Alteryx, and UiPath RPA) that digitized 100% of the preparer work for the Evidence Gathering Audit (EGA) process and automated uploading of EGAs to the internal audit portal, saving hundreds of hours across the Financial Services cluster.
- **Project Management Tracker:** Programmed a robust Alteryx ETL workflow integrated with Google Sheets APIs to extract, transform, and load live project tracking metrics directly into a Tableau dashboard, standardizing and automating multi-source reporting.
- **Google Sheet to Web App Conversion:** Developed and deployed a full-stack Google Apps Script Web Application (HTML5, CSS3, JavaScript) to standardize user inputs and enforce real-time data validation, eliminating downstream RPA bot failures caused by malformed data.
- Deployed Excel VBA solutions to speed up data preprocessing, reporting, and auditing processes in HR, Learning & Development, and Risk & Quality.
- Pioneered the Data Analytics and Visualization delivery team, establishing the first team to deliver chargeable VBA, Alteryx, and Power BI solutions across HK, SG, and AU regions.
- Promoted from Associate to Analyst role (2020) and transitioned from Contractual to FTE (2018).

Trainings:

- Alteryx Designer Core Certified (2019)
- DOST-Coursera Grant (2020, 2018)

EDUCATION

B.S. Industrial Engineering

2012 – 2017

Polytechnic University of the Philippines — Sta. Mesa, Manila

Civil Service Professional Passer · Civil Service SubProfessional Passer · DOST-SEI Scholar (2014)

M.S. Engineering, Major in Computer Engineering — 15 units earned

2017 – 2019

Polytechnic University of the Philippines

DOST-PCIEERD Data, Connectivity and Intelligence: Data Science Track Scholar 2017 Recipient (Discontinued)

Post Baccalaureate Diploma in Information Technology — 9 units earned

2017 – 2019

Polytechnic University of the Philippines (in parallel with Masters, Discontinued)

RESEARCH INTERESTS

My research interests lie at the intersection of **Operations Research, Machine Learning, and Decision Science** — with a focus on formulating real-world business and operational challenges as tractable optimization problems:

- ▶ **Resource Optimization & Scheduling:** Workforce capacity planning, staffing allocation under uncertainty, and multi-objective scheduling formulations (LP/MIP)
- ▶ **Queueing-Theoretic Modeling:** Analytical and simulation-based approaches to service system design and capacity dimensioning
- ▶ **Explainable ML for Decision Systems:** Interpretable predictive models (SHAP/LIME) for high-stakes domains such as financial services and customer retention
- ▶ **AI-Driven Process Optimization:** Integrating ML predictions with prescriptive optimization to close the loop from insight to action

TECHNICAL SKILLS

📐 Operations Research & Optimization

Mathematical Optimization: LP, MIP, Constraint Programming (MiniZinc, Excel Solver, R, Python) · Queueing Theory & Simulation · Resource Allocation, Scheduling & Capacity Planning · Design of Experiments & A/B Testing (Python, R, Excel, Minitab) · Statistical Analysis/Hypothesis Testing & EDA

⚙️ Data Engineering & ML Pipelines

Data Extraction (API, Databases, Web Scraping) · SQL Scripting & Querying · ETL Pipelines (Python/Pandas/Polars, Alteryx, PowerShell, Batch) · End-to-end ML pipeline design · Dashboard & Visualization (Tableau, Power BI, D3.js)

🧠 Machine Learning & Statistical Modeling

Regression & Classification (Python, R) · Predictive & Prescriptive Analysis · Forecasting & Time Series (ARIMA, SARIMAX, Prophet) · Survival Analysis · Explainable AI (SHAP, LIME) · NLP & Transformer Models (BERT via HuggingFace) · AutoML (PyCaret, AutoGluon) · Business metric/OKR tracking for DS projects

🤖 AI Agents & Automation

RAG Pipeline Architecture · LLM Agent Development · MS365 Copilot · Power Automate · Excel VBA · UiPath RPA · PowerQuery & DAX · Google Apps Script · HTML/CSS/JavaScript

CERTIFICATIONS & COURSES

Machine Learning & AI

- ▶ Analyze Datasets and Train ML Models using AutoML — Coursera
- ▶ Build, Train, and Deploy ML Pipelines using BERT — Coursera

Data Science & Statistics

- ▶ Experimental Design Basics — Coursera
- ▶ Data Science: Foundation using R — Coursera Specialization
- ▶ Data Analysis & Presentation Skills: the PwC Approach — Coursera
- ▶ Python for Time Series Data Analysis — Udemy

Python Programming

- ▶ Python for Everybody — Coursera Specialization
- ▶ Python 3 Programming — Coursera Specialization

Professional Certifications

- ▶ COPC Lean Six Sigma Yellow Belt (2022)
- ▶ Alteryx Designer Core Certified (2019)
- ▶ Project Management Fundamentals — PMI
- ▶ Data Storytelling for Business — StoryIQ

AWARDS & ACHIEVEMENTS

- **Best Data Scientist Award** — Teleperformance Philippines (2023)
- **TP Datathon — 1st Place (Regional)** — Won first place in the regional data science and analytics hackathon (2021)
- **TP Datathon — 3rd Place (Global)** — Placed third globally among international teams, designing and presenting predictive models (2021)

ACADEMIC TALKS & PANELIST ROLES

2025

- IEAC 2025 Alumni Guest Speaker — PUP Sta. Mesa, Industrial Engineering Department (June 2025)

2023

- IE4DS: IE's Competitive Advantage, Challenges and Opportunities for IEs in Data Science — PUP Sta. Mesa, Industrial Engineering Department (July 2023)

2022

- Getting Started with Data Analytics — Asia Technological School of Science and Arts, College of Engineering and IT Education (December 2022)
- KnowledgeED Industries: Upskilling for a Data Science Job as an IE Student — PIIE LPU Cavite (February 2022)

2021

- Data Mining: Process, Concept and Tools — PUP ACCESS, Computer Engineering Department (September 2021)
- Prescriptive Analytics: Excel, R and Python for Solving Linear Programming Problems — PIIE LPU Cavite Student Chapter (June 2021)
- IE As a Data Analyst and Automation Developer — Industrial Engineering Department, Quezon City University (February 2021)

2020

- Solving Real World Problems in OR — Industrial Engineering Department, Quezon City University (November 2020)
- Excel Beyond the Basics (Advanced Formulas, Data Analysis, Solver, VBA) — Industrial Engineering Department, TIP-QC (September 2020)
- Automation with Excel VBA — ALPHA TV Webinar | Ask Lex PH Academy (May 2020)
- Data Analysis and Visualization — Industrial Engineering Annual Convention 2020 | PIIE PUP Sta. Mesa (February 2020)
- Optimization: From Data Science and Operations Research — Industrial Engineering Symposium 2020 | IE Department, PUP Biñan Campus (February 2020)
- Introduction to Data Science and Machine Learning with Python — ECE Special Lecture Series 2020 | ECE Department, PUP Sta. Rosa Campus (February 2020)

2019

- Prescriptive Analytics: Simulation using Excel's What-if Analysis — IEShikawa 14 by UP-CIEM | University of the Philippines - Diliman (November 2019)
- Move Forward in AI: OR in AI — ORSP-UST Chapter | University of Sto. Tomas (November 2019)
- Research Panelist, Science Research Fair and Congress 2019 — Antipolo City National Science and Technology High School (March 2019)

INTERNSHIPS

Toyota Makati Inc. — Kaizen Intern (320 hrs)

2016

Applied Toyota Business Practice framework, Kaizen, and Lean Manufacturing principles to map processes, analyze data, and visualize root causes in the Parts Department warehouse.

Khumbu Enterprise Inc. (Sandugo) — Work Study Intern (300 hrs)

2015

Conducted time and motion study under Merchandising Department's logistics warehouse to evaluate staffing needs and improve work methods.

VST ECS Phils., Inc. — Work Study Intern (72 hrs)

2015

Created detailed documentation of the process flow in the provincial shipping section of the warehouse for process standardization.

PERSONAL PROJECTS

QueuePro: Industrial Queueing Simulator

[Queueing Theory](#) · [Discrete Event Simulation](#) · [Stochastic Modeling](#) · [SAAS](#)

QueuePro is a professional Discrete Event Simulation (DES) SAAS platform designed for Process Engineers, Data Scientists, and Educators. Optimize your service floor, eliminate bottlenecks, and validate system capacity using rigorous stochastic modeling.

Room Reservation App for Local Church

[HTML](#) · [CSS](#) · [JavaScript](#) · [Google Apps Script](#)

A room reservation application that automates the booking process for church rooms, allowing authorized users to reserve rooms for specific time slots with double-booking prevention.

Brilliant.org Problem Solver

[Math](#) · [Statistics](#) · [Computer Science](#)

Actively solving Math, Statistics, and Computer Science quizzes on Brilliant.org for continuous learning.

Personal Math Tutor

[Geometry](#) · [Trigonometry](#) · [Algebra](#) · [Pre-Calculus](#)

Personal math tutoring for high school students.

Personal Interactive Resume

[HTML](#) · [CSS](#) · [JavaScript](#) · [GitHub Pages](#)

A comprehensive, interactive web-based portfolio and resume hosted on GitHub Pages: [View here](#).